

Практика 1

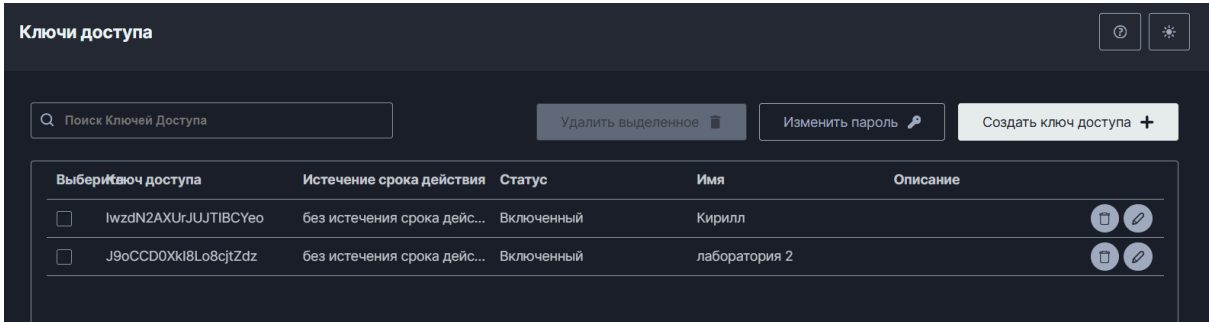
группа: "Управление на основе данных"

Выполнил: Гусев Кирилл

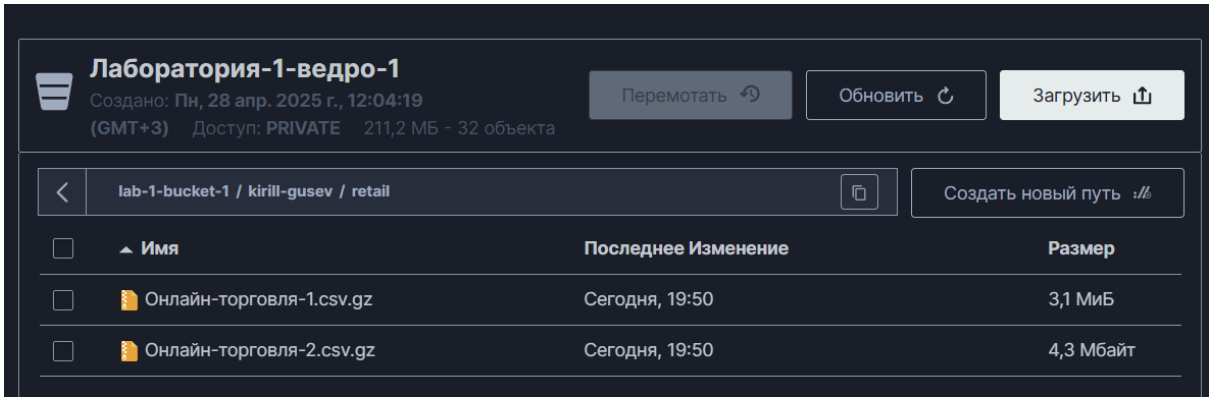
Тема: Использование кластера ClickHouse

Задание 1. Добавление файлов в объектное хранилище (S3)

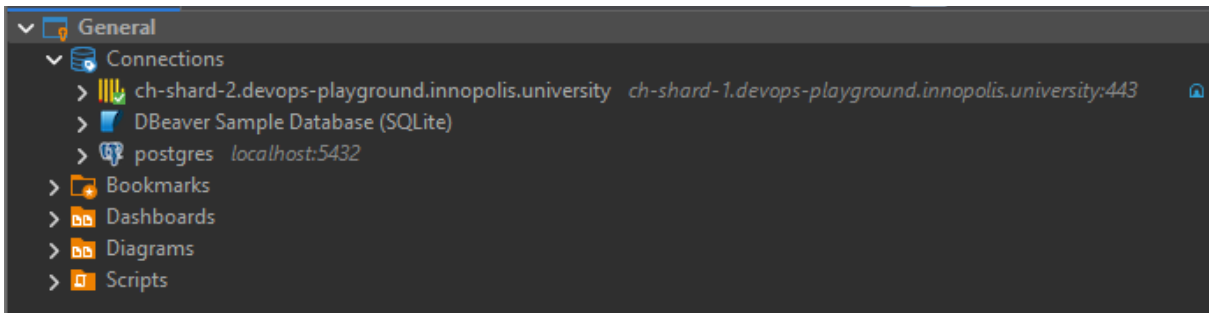
1.



2.



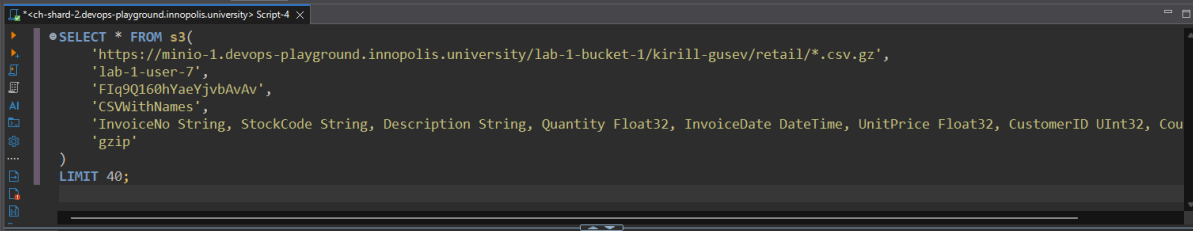
3.



В первом этапе было выполнено следующее:

- создание ключей доступа в Minio
- загрузка файлов в S3
- Подключение к ClickHouse

Задание 2.Чтение данных из S3 в Clickhouse



The screenshot shows a ClickHouse SQL client window with a query and its results. The query is:

```
SELECT * FROM s3(
  'https://minio-1.devops-playground.innopolis.university/lab-1-bucket-1/kirill-gusev/retail/*.csv.gz',
  'lab-1-user-7',
  'FIq9Q160hYaeYjvbAvAv',
  'CSVWithNames',
  'InvoiceNo String, StockCode String, Description String, Quantity Float32, InvoiceDate DateTime, UnitPrice Float32, CustomerID UInt32, Country String'
)
LIMIT 40;
```

The results are displayed in a table with the following columns: InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, and Country. The table contains 40 rows of data, showing various products and their quantities, prices, and countries.

InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	2010-12-01 11:26:00	2.55	17 850	United Kingdom
536365	71053	WHITE METAL LANTERN	6	2010-12-01 11:26:00	3.3899999	17 850	United Kingdom
536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	2010-12-01 11:26:00	2.75	17 850	United Kingdom
536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	2010-12-01 11:26:00	3.3899999	17 850	United Kingdom
536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	2010-12-01 11:26:00	3.3899999	17 850	United Kingdom
536365	22752	SET 7 BABUSHKA NESTING BOXES	2	2010-12-01 11:26:00	7.65	17 850	United Kingdom
536365	21730	GLASS STAR FROSTED T-LIGHT HOLDER	6	2010-12-01 11:26:00	4.25	17 850	United Kingdom
536366	22633	HAND WARMER UNION JACK	6	2010-12-01 11:28:00	1.85	17 850	United Kingdom
536366	22632	HAND WARMER RED POLKA DOT	6	2010-12-01 11:28:00	1.85	17 850	United Kingdom
536367	84879	ASSORTED COLOUR BIRD ORNAMENT	32	2010-12-01 11:34:00	1.69	13 047	United Kingdom
536367	22745	POPPY'S PLAYHOUSE BEDROOM	6	2010-12-01 11:34:00	2.1	13 047	United Kingdom
536367	22748	POPPY'S PLAYHOUSE KITCHEN	6	2010-12-01 11:34:00	2.1	13 047	United Kingdom
536367	22740	FELTCRAFT PRINCESS CHARLOTTE DOLL	8	2010-12-01 11:34:00	3.75	13 047	United Kingdom
536367	22310	IVORY KNITTED MUG COSY	6	2010-12-01 11:34:00	1.65	13 047	United Kingdom
536367	84969	BOX OF 6 ASSORTED COLOUR TEASPOONS	6	2010-12-01 11:34:00	4.25	13 047	United Kingdom
536367	22623	BOX OF VINTAGE JIGSAW BLOCKS	3	2010-12-01 11:34:00	4.95	13 047	United Kingdom
536367	22622	BOX OF VINTAGE ALPHABET BLOCKS	2	2010-12-01 11:34:00	9.95	13 047	United Kingdom
536367	21754	HOME BUILDING BLOCK WORD	3	2010-12-01 11:34:00	5.95	13 047	United Kingdom
536367	21755	LOVE BUILDING BLOCK WORD	3	2010-12-01 11:34:00	5.95	13 047	United Kingdom
536367	21777	RECIPE BOX WITH METAL HEART	4	2010-12-01 11:34:00	7.95	13 047	United Kingdom
536367	48187	DOORMAT NEW ENGLAND	4	2010-12-01 11:34:00	7.95	13 047	United Kingdom
536368	22960	JAM MAKING SET WITH JARS	6	2010-12-01 11:34:00	4.25	13 047	United Kingdom
536368	22913	RED COAT RACK PARIS FASHION	3	2010-12-01 11:34:00	4.95	13 047	United Kingdom
536368	22912	YELLOW COAT RACK PARIS FASHION	3	2010-12-01 11:34:00	4.95	13 047	United Kingdom
536368	22914	BLUE COAT RACK PARIS FASHION	3	2010-12-01 11:34:00	4.95	13 047	United Kingdom
536369	21756	BATH BUILDING BLOCK WORD	3	2010-12-01 11:35:00	5.95	13 047	United Kingdom
536370	22728	ALARM CLOCK BAKELIKE PINK	24	2010-12-01 11:45:00	3.75	12 583	France
536370	22727	ALARM CLOCK BAKELIKE RED	24	2010-12-01 11:45:00	3.75	12 583	France
536370	22726	ALARM CLOCK BAKELIKE GREEN	12	2010-12-01 11:45:00	3.75	12 583	France
536370	21724	PANDA AND BUNNIES STICKER SHEET	12	2010-12-01 11:45:00	0.85	12 583	France
536370	21883	STARS GIFT TAPE	24	2010-12-01 11:45:00	0.65	12 583	France
536370	10002	INFLATABLE POLITICAL GLOBE	48	2010-12-01 11:45:00	0.85	12 583	France
536370	21791	VINTAGE HEADS AND TAILS CARD GAME	24	2010-12-01 11:45:00	1.25	12 583	France
536370	21035	SET/2 RED RETROSPOT TEA TOWELS	18	2010-12-01 11:45:00	2.95	12 583	France
536370	22326	ROUND SNACK BOXES SET OF4 WOODLAND	24	2010-12-01 11:45:00	2.95	12 583	France
536370	22629	SPACEBOY LUNCH BOX	24	2010-12-01 11:45:00	1.95	12 583	France
536370	22659	LUNCH BOX I LOVE LONDON	24	2010-12-01 11:45:00	1.95	12 583	France
536370	22631	CIRCUS PARADE LUNCH BOX	24	2010-12-01 11:45:00	1.95	12 583	France
536370	22661	CHARLOTTE BAG DOLLY GIRL DESIGN	20	2010-12-01 11:45:00	0.85	12 583	France
536370	21731	RED TOADSTOOL LED NIGHT LIGHT	24	2010-12-01 11:45:00	1.65	12 583	France

```
SELECT * FROM s3(
  'https://minio-1.devops-playground.innopolis.university/lab-1-bucket-1/kirill-gusev/retail/*.csv.gz',
  'lab-1-user-7',
  'FIq9Q160hYaeYjvbAvAv',
```

```

    'CSVWithNames',
    'InvoiceNo String, StockCode String, Description
String, Quantity Float32, InvoiceDate DateTime,
UnitPrice Float32, CustomerID UInt32, Country
String',
    'gzip'
)
LIMIT 40;

```

Был выполнен скрипт из презентации, в результате чего
выполнилось подключение и чтение данных из s3 через
Clickhouse.

**Задание 3. Использование внешней таблицы вместо
запроса**

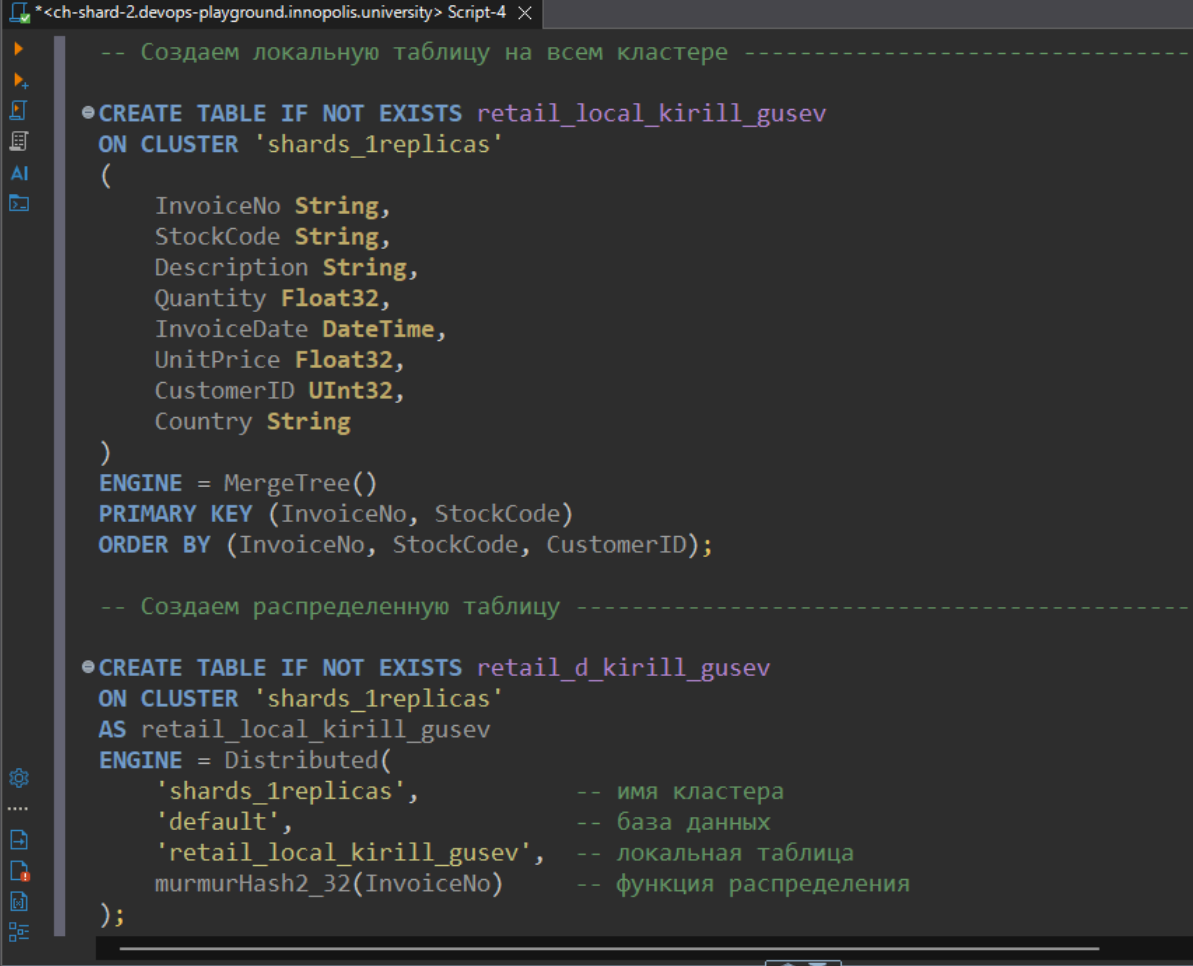
```

CREATE TABLE retail_kirill_gusev_s3 (
    InvoiceNo String,
    StockCode String,
    Description String,
    Quantity Float32,
    InvoiceDate DateTime,
    UnitPrice Float32,
    CustomerID UInt32,
    Country String
)
ENGINE = S3
('https://minio-1.devops-playground.innopolis.univers
ity/lab-1-bucket-1/kirill-gusev/retail/Online_Retail-
1.csv',
    'lab-1-user-7',
    'FIq9Q160hYaeYjvbAvAv',
    'CSVWithNames',
    'none'
);

```

В результате выполнения скрипта была создана таблица

Задание 4. Подготовка таблиц для хранения данных



```
-- Создаем локальную таблицу на всем кластере -----
CREATE TABLE IF NOT EXISTS retail_local_kirill_gusev
ON CLUSTER 'shards_1replicas'
(
    InvoiceNo String,
    StockCode String,
    Description String,
    Quantity Float32,
    InvoiceDate DateTime,
    UnitPrice Float32,
    CustomerID UInt32,
    Country String
)
ENGINE = MergeTree()
PRIMARY KEY (InvoiceNo, StockCode)
ORDER BY (InvoiceNo, StockCode, CustomerID);

-- Создаем распределенную таблицу -----
CREATE TABLE IF NOT EXISTS retail_d_kirill_gusev
ON CLUSTER 'shards_1replicas'
AS retail_local_kirill_gusev
ENGINE = Distributed(
    'shards_1replicas',           -- имя кластера
    'default',                   -- база данных
    'retail_local_kirill_gusev',  -- локальная таблица
    murmurHash2_32(InvoiceNo)     -- функция распределения
);
```

unknown 1 ×

CREATE TABLE IF NOT EXISTS retail_d_kirill_gusev | Введите SQL выражение чтобы отфильтровать результаты

	AZ host	123 port	123 status	AZ error	123 num_hosts_remaining	123 num_hosts_active
1	clickhouse-shard-1	9 000	0		1	0
2	clickhouse-shard-2	9 000	0		0	0

```
-- Создаем локальную таблицу на всем кластере

CREATE TABLE IF NOT EXISTS retail_local_kirill_gusev
```

```

ON CLUSTER 'shards_1replicas'
(
    InvoiceNo String,
    StockCode String,
    Description String,
    Quantity Float32,
    InvoiceDate DateTime,
    UnitPrice Float32,
    CustomerID UInt32,
    Country String
)
ENGINE = MergeTree()
PRIMARY KEY (InvoiceNo, StockCode)
ORDER BY (InvoiceNo, StockCode, CustomerID);

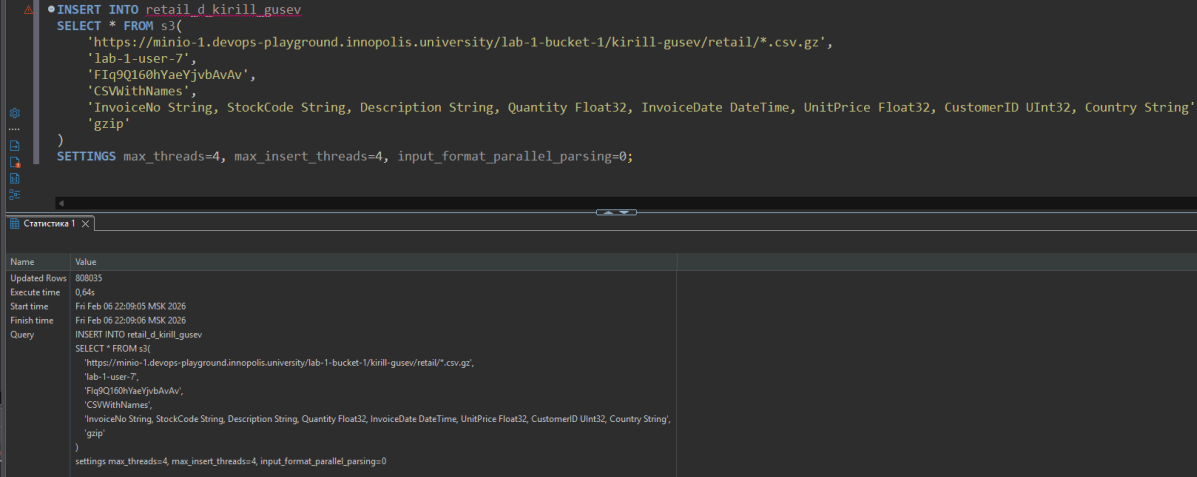
-- Создаем распределенную таблицу

CREATE TABLE IF NOT EXISTS retail_d_kirill_gusev
ON CLUSTER 'shards_1replicas'
AS retail_local_kirill_gusev
ENGINE = Distributed(
    'shards_1replicas',           -- имя кластера
    'default',                   -- база данных
    'retail_local_kirill_gusev',  -- локальная таблица
    murmurHash2_32(InvoiceNo)     -- функция
распределения
);

```

На данном этапе была создана таблица на всем кластере а также была создана распределенная таблица

Задание 5. Заполнение распределенной таблицы из файлов в S3



Name	Value
Updated Rows	808035
Execute time	0.64s
Start time	Fri Feb 06 22:09:05 MSK 2020
Finish time	Fri Feb 06 22:09:06 MSK 2020
Query	INSERT INTO retail_d_kirill_gusev SELECT * FROM s3('https://minio-1.devops-playground.innopolis.university/lab-1-bucket-1/kirill-gusev/retail/*.csv.gz', 'lab-1-user-7', 'FIq9Q160hYaeYjvbAvAv', 'CSVWithNames', 'InvoiceNo String, StockCode String, Description String, Quantity Float32, InvoiceDate DateTime, UnitPrice Float32, CustomerID UInt32, Country String', 'gzip') SETTINGS max_threads=4, max_insert_threads=4, input_format_parallel_parsing=0;

```
INSERT INTO retail_d_kirill_gusev
SELECT * FROM s3(

'https://minio-1.devops-playground.innopolis.universi
ty/lab-1-bucket-1/kirill-gusev/retail/*.csv.gz',
  'lab-1-user-7',
  'FIq9Q160hYaeYjvbAvAv',
  'CSVWithNames',
  'InvoiceNo String, StockCode String, Description
String, Quantity Float32, InvoiceDate DateTime,
UnitPrice Float32, CustomerID UInt32, Country
String',
  'gzip'
)
SETTINGS max_threads=4, max_insert_threads=4,
input_format_parallel_parsing=0;
```

Задание 6. Напишите запрос select from where для фильтрации данных

```
SELECT
    CustomerID,
    COUNT(DISTINCT InvoiceNo) as order_count,
    SUM(Quantity) as total_items,
    SUM(Quantity * UnitPrice) as total_spent,
    MIN(InvoiceDate) as first_purchase,
    MAX(InvoiceDate) as last_purchase
FROM retail_d_kirill_gusev
WHERE Description LIKE '%HEART%'
GROUP BY CustomerID
HAVING total_spent > 0
ORDER BY total_spent DESC
LIMIT 10;
```

	123 CustomerID	123 order_count	123 total_items	123 total_spent	first_purchase	last_purchase
1	15 287	864	67 718	207 373,7203029394	2010-12-01 17:32:00	2011-12-09 14:26:00
2	17 450	24	35 472	107 162,4409580231	2010-12-07 12:43:00	2011-11-29 14:16:00
3	18 102	39	33 806	101 299,1985576153	2010-12-07 19:42:00	2011-12-08 22:43:00
4	14 646	33	34 062	44 505,299836278	2010-12-21 18:05:00	2011-12-07 19:19:00
5	14 911	144	14 316	21 525,83994174	2010-12-01 17:05:00	2011-12-08 19:54:00
6	13 777	25	7 374	17 679,9797611237	2010-12-01 19:01:00	2011-12-09 16:25:00
7	14 156	37	8 404	17 385,6398899555	2011-01-14 17:11:00	2011-11-30 14:54:00
8	14 298	24	17 472	16 658,51980865	2010-12-14 15:59:00	2011-11-09 14:26:00
9	13 089	50	8 800	14 943,040012598	2010-12-06 16:59:00	2011-12-05 18:30:00
10	16 013	16	4 062	11 819,8597521782	2010-12-16 14:07:00	2011-12-02 15:53:00

```
SELECT
    CustomerID,
    COUNT(DISTINCT InvoiceNo) as order_count,
    SUM(Quantity) as total_items,
    SUM(Quantity * UnitPrice) as total_spent,
    MIN(InvoiceDate) as first_purchase,
    MAX(InvoiceDate) as last_purchase
FROM retail_d_kirill_gusev
WHERE Description LIKE '%HEART%'
GROUP BY CustomerID
HAVING total_spent > 0
ORDER BY total_spent DESC
LIMIT 10;
```

Задание 7. Напишите ещё один запрос, который может быть использован бизнес-аналитиком

Проведем анализ продаж по странам:

```
SELECT
    'Анализ продаж по странам' as report_title,
    Country,
    COUNT(DISTINCT CustomerID) as unique_customers,
    COUNT(DISTINCT InvoiceNo) as total_orders,
    SUM(Quantity) as items_sold,
    ROUND(SUM(Quantity * UnitPrice), 2) as total_revenue,
    ROUND(total_revenue / COUNT(DISTINCT InvoiceNo), 2) as avg_order_value,
    ROUND(total_revenue / COUNT(DISTINCT CustomerID), 2) as revenue_per_customer
FROM retail_d_kirill_gusev
GROUP BY Country
ORDER BY total_revenue DESC
LIMIT 15;
```

	A1 report_title	A2 Country	I23 unique_customers	I23 total_orders	I23 items_sold	I23 total_revenue	I23 avg_order_value	I23 revenue_per_customer
1	Анализ продаж по странам	United Kingdom	3 950	23 494	8 528 623	16 378 152,74	697,12	4 146,37
2	Анализ продаж по странам	Netherlands	9	101	400 256	5 636,86	5 636,86	63 258,12
3	Анализ продаж по странам	EIRE	4	360	285 274	526 553,64	1 462,65	131 638,41
4	Анализ продаж по странам	Germany	135	643	235 156	457 166,42	710,99	3 386,42
5	Анализ продаж по странам	France	128	501	221 709	411 813,66	821,98	3 217,29
6	Анализ продаж по странам	Australia	9	69	167 306	274 154,54	3 973,25	30 461,62
7	Анализ продаж по странам	Switzerland	22	74	60 650	112 770,7	1 523,93	5 125,94
8	Анализ продаж по странам	Spain	31	105	53 648	109 549,16	1 043,33	3 533,84
9	Анализ продаж по странам	Belgium	25	119	46 304	81 821,92	687,58	3 272,88
10	Анализ продаж по странам	Sweden	8	46	71 274	73 191,82	1 591,13	9 148,98
11	Анализ продаж по странам	Japan	8	28	50 436	70 681,24	2 524,33	8 835,16
12	Анализ продаж по странам	Norway	10	40	38 494	70 326,92	1 758,17	7 032,69
13	Анализ продаж по странам	Portugal	20	71	32 360	58 734,04	827,24	2 986,7
14	Анализ продаж по странам	Finland	12	48	21 332	44 653,48	930,28	3 721,12
15	Анализ продаж по странам	Channel Islands	9	33	18 958	40 172,58	1 217,35	4 463,62

```
SELECT
    'Анализ продаж по странам' as report_title,
    Country,
    COUNT(DISTINCT CustomerID) as unique_customers,
    COUNT(DISTINCT InvoiceNo) as total_orders,
    SUM(Quantity) as items_sold,
    ROUND(SUM(Quantity * UnitPrice), 2) as
total_revenue,
    ROUND(total_revenue / COUNT(DISTINCT InvoiceNo),
2) as avg_order_value,
    ROUND(total_revenue / COUNT(DISTINCT CustomerID),
2) as revenue_per_customer
FROM retail_d_kirill_gusev
GROUP BY Country
ORDER BY total_revenue DESC
LIMIT 15;
```